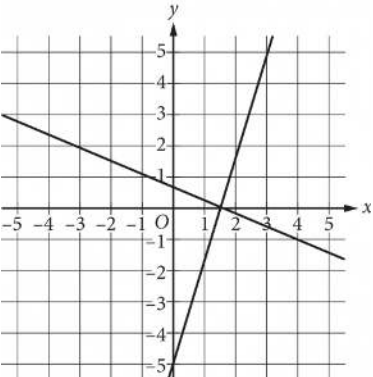


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question



Which of the following systems of equations has the same solution as the system of equations graphed above?

Answer

- $y = 0$
- A. $x = \frac{3}{2}$
- B. $y = \frac{3}{2}$
 $x = 0$
- C. $y = 0$
 $x = 1$
- D. $y = 1$
 $x = 0$

Correct Answer: A

Rationale

Choice A is correct. The solution to a system of equations is the coordinates of the intersection point of the graphs of the equations in the xy-plane. Based on the graph, the solution to the given system of equations is best approximated as $(\frac{3}{2}, 0)$. In the xy-plane, the graph of $y = 0$ is a horizontal line on which every y-coordinate is 0, and the graph of $x = \frac{3}{2}$ is a vertical line on which every x-coordinate is $\frac{3}{2}$. These graphs intersect at the point $(\frac{3}{2}, 0)$. Therefore, the system of equations in choice A has the same solution as the given system.

Choices B, C, and D are incorrect. If graphed in the xy -plane, these choices would intersect at the points $(0, \frac{3}{2})$, $(1, 0)$, and $(0, 1)$, respectively,

not $(\frac{3}{2}, 0)$.